



Online and Reinforcement Learning

ID6002W

P Satya Jayadev

Web Enabled M.Tech,
Indian Institute of Technology Madras

Where We Are in the Course

- ▶ The first part of the course focused on **Online Learning**.
- ▶ You studied how a learner can make repeated decisions and improve using feedback.



Where We Are in the Course

- ▶ The first part of the course focused on **Online Learning**.
- ▶ You studied how a learner can make repeated decisions and improve using feedback.
- ▶ Key ideas already introduced:
 - Actions / arms
 - Rewards
 - Exploration vs. exploitation
 - Cumulative reward and regret
 - Contextual and structured bandit settings



Where We Are in the Course

- ▶ The first part of the course focused on **Online Learning**.
- ▶ You studied how a learner can make repeated decisions and improve using feedback.
- ▶ Key ideas already introduced:
 - Actions / arms
 - Rewards
 - Exploration vs. exploitation
 - Cumulative reward and regret
 - Contextual and structured bandit settings
- ▶ In this part, we move from **online decisions** to **sequential decision-making**.

Online Learning $\longrightarrow$ **Reinforcement Learning**



Outline of RL Part

1. Intro to RL
2. Markov Decision Processes (MDPs)
3. Monte Carlo Learning Methods
4. Temporal Difference Learning Methods
5. Value Function Approximation based Methods
6. Policy Gradient Methods
7. Actor Critic Methods
8. Deep Reinforcement Learning
9. Modern Advances in RL



1. Text Book: Reinforcement Learning: An Introduction by Sutton, R. S., Barto, A. G. (2018) - PDF Link
2. Reinforcement Learning : An Overview, Kevin Murphy PDF Link
3. Recorded Lectures by Prof. B. Ravindran: Youtube Playlist
4. Recorded Lectures by Dr. David Silver: Youtube Playlist



Evaluation Plan for RL Part

- ▶ The **Reinforcement Learning part** will be evaluated for **50 marks**.
- ▶ Evaluation will be based on:
 - **Three assignments (Objective Programming): 35%**
 - **Course project with competitive leaderboard: 65%**

Note

There will be **no separate final exam** for the RL part. The final component will be a project evaluated through implementation, analysis, leaderboard performance and project report.



Why Reinforcement Learning?

Motivating RL from SL perspective

Supervised learning helps us predict.

Reinforcement learning helps us act optimally.

Where Supervised and Unsupervised Learning Are Not Enough

- ▶ Many practical problems involve **decisions**, not just predictions.
- ▶ The correct action may not be known in advance.
- ▶ The learner may need to try different actions to discover what works.
- ▶ A good immediate decision may not always lead to the best long-term outcome.



Where Supervised and Unsupervised Learning Are Not Enough

- ▶ Many practical problems involve **decisions**, not just predictions.
- ▶ The correct action may not be known in advance.
- ▶ The learner may need to try different actions to discover what works.
- ▶ A good immediate decision may not always lead to the best long-term outcome.

Example	Why SL/UL may be inadequate
Game playing	Correct move is not labelled for every possible board position
Autonomous driving	Actions have future consequences and safety implications
Industrial process control	Current control action affects future process behaviour
Inventory management	Ordering today affects future stock, cost, and service level



Supervised Learning vs Reinforcement Learning

Aspect	Supervised Learning	Reinforcement Learning
Training signal	Correct labels or target values	Rewards from the environment
Data source	Fixed dataset	Experience generated through interaction
Learner output	Prediction model	Decision-making policy
Objective	Minimize prediction error	Maximize long-term reward
Main challenge	Generalization to unseen data	Learning actions that improve long-term outcomes

Key Idea

Reinforcement Learning is useful when we do not have labelled examples of the correct action, but we can observe the consequences of actions.



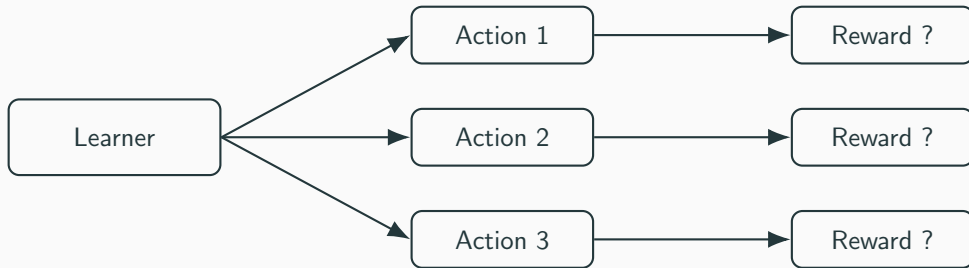
From Online Learning to Reinforcement Learning

What carries forward, and what changes?

Online learning taught us how to learn from feedback.

Reinforcement learning adds future consequences.

Recall: Online Learning / Bandit Framework



- ▶ The learner repeatedly chooses among available actions.
- ▶ It does not know in advance which action will work best.
- ▶ It learns from reward feedback and improves future choices.
- ▶ **Exploration**: trying less-known actions to learn more.
- ▶ **Exploitation**: using actions that seem to work well.



From Online Learning Ideas to RL Challenges

- Reinforcement Learning builds on the ideas from online learning.
- But each idea becomes more challenging because actions can affect future situations.

In Online Learning / Bandits	What Changes in RL
Choose among actions or arms	Choose actions depending on the current situation
Reward is observed after an action	Reward may be delayed or spread over many future steps
Exploration helps identify better actions	Exploration can change future situations and future learning opportunities
Goal is high cumulative reward or low regret	Goal is good long-term behaviour across a sequence of situations
Past data improves future choices	Past experience is used to improve how the agent behaves in different states



Why Future Consequences Matter

- ▶ In RL, the quality of an action cannot always be judged immediately.
- ▶ An action may look good in the short term, but create problems later.
- ▶ Similarly, an action may look costly now, but lead to better outcomes later.



Why Future Consequences Matter

- ▶ In RL, the quality of an action cannot always be judged immediately.
- ▶ An action may look good in the short term, but create problems later.
- ▶ Similarly, an action may look costly now, but lead to better outcomes later.
- ▶ RL is useful when decisions must be evaluated by their effect on future outcomes, not just immediate feedback.



Why Future Consequences Matter

- ▶ In RL, the quality of an action cannot always be judged immediately.
- ▶ An action may look good in the short term, but create problems later.
- ▶ Similarly, an action may look costly now, but lead to better outcomes later.
- ▶ RL is useful when decisions must be evaluated by their effect on future outcomes, not just immediate feedback.

Example Situation	Immediate View	Long-Term View
Delay machine maintenance	Saves downtime today	May increase breakdown risk later
Order less inventory	Reduces holding cost now	May cause stockouts later
Increase heating/fuel intensity in a furnace	May reach target faster	May increase energy cost or affect quality
Accept too many orders	Improves short-term utilization	May increase delays and penalties



The Missing Ingredient: State

- ▶ To reason about future consequences, the learner needs to know the current situation.
- ▶ In RL, this current situation is called the **state**.



The Missing Ingredient: State

- ▶ To reason about future consequences, the learner needs to know the current situation.
- ▶ In RL, this current situation is called the **state**.
- ▶ The same action may be good in one state and poor in another.



The Missing Ingredient: State

- ▶ To reason about future consequences, the learner needs to know the current situation.
- ▶ In RL, this current situation is called the **state**.
- ▶ The same action may be good in one state and poor in another.
- ▶ So RL is not only about choosing the best action overall.
- ▶ It is about choosing the right action for the current situation.

Current State $\longrightarrow$ **Action** $\longrightarrow$ **Outcome** $\longrightarrow$ **New State**

Problem	Examples of State
Inventory control	Current stock, pending orders, demand trend
Machine maintenance	Machine age, vibration level, temperature, past failures
Production scheduling	Backlog, machine availability, due dates, work-in-progress



Reinforcement Learning: Learning by Interaction

- ▶ Similar to how humans learn to ride a bicycle or ride a car
- ▶ A learning **agent** interacts with an **environment** through a sequence of decisions or actions



Reinforcement Learning: Learning by Interaction

- ▶ Similar to how humans learn to ride a bicycle or ride a car
- ▶ A learning **agent** interacts with an **environment** through a sequence of decisions or actions
- ▶ At each step:
 - Agent observes the environment's **state**
 - Takes an **action**
 - Receives feedback called **reward**
 - Environment changes state



Reinforcement Learning: Learning by Interaction

- ▶ Similar to how humans learn to ride a bicycle or ride a car
- ▶ A learning **agent** interacts with an **environment** through a sequence of decisions or actions
- ▶ At each step:
 - Agent observes the environment's **state**
 - Takes an **action**
 - Receives feedback called **reward**
 - Environment changes state
- ▶ **Learning Goal:** To learn a strategy called **policy** that maximizes cumulative reward called **return**
- ▶ This interaction will be formalised using a Markov Decision Processes.

